The J_1–J_2 model on $4 \times L_x$ cylinders: inputs, trial states, costs and crossover

Data report for the filter-pipeline paper

October 28, 2026

This report collects every two-dimensional result produced for the paper. All tables and figures are generated by `sbpipe_study/report2d/make_report2d.py` from `sbpipe_study/results/*.json` (`ext_inputs2d.json`, `cache2d/`, `ext_regime_cyl*.json`, `emp_regime.json`, `emp_frontier.json`). Nothing in it is typed by hand.

1 Model and pipeline settings

$$H = J_1 \sum_{\langle ij \rangle} \mathbf{S}_i \cdot \mathbf{S}_j + J_2 \sum_{\langle\langle ij \rangle\rangle} \mathbf{S}_i \cdot \mathbf{S}_j, \qquad J_1 = 1, \tag{1}$$

on an $L_x \times 4$ square lattice, periodic around the four-site circumference and open along $L_x = 3, 4, 6, 8, 12$ ($N = 12$–48 qubits), at $J_2 = 0$ and 0.3. Site (x, y) is qubit $4x + y$, so the MPS runs up each column in turn (a snake path). $J_2 = 0.5$ is left out because ED on 4×4 shows that its lowest excitation is not the triplet, while the pipeline takes E_1 as the lowest $S^z = 1$ state.

- **Energies.** DMRG (quimb) with reference bond dimensions $\chi = 16, \ldots, 256$. E_0 and E_1 (the lowest $S^z = 1$ state) are taken from $\chi = 256$, with the error estimated from the last change in χ. The spectral upper bound is Weyl's. ED is run for 4×3 and 4×4.
- **Trial states.** Variational MPS at $\chi = 2, \ldots, 64$. The overlap $\gamma = |\langle E_0 | \psi_\chi \rangle|^2$ is measured against the $\chi = 256$ reference (or ED). Each trial is costed as a staircase of quantum-Shannon-decomposed unitaries.
- **Filter.** QETU (Chebyshev) or the dyadic/optimized cosine filters, second-order Strang steps, CNOT counts from Pauli-gadget ladders. *Certified* costs come from the pipeline's a-priori bound. *Empirical* costs use the two laws measured on exact spectra for $N \leq 16$ (675 cases): the smallest working degree is $d\,\Delta \simeq (2/\pi) \cosh^{-1}(1/\eta)$ with $\eta^2 = \gamma\epsilon/(1-\gamma)$, and one Strang step per query suffices.

2 Energies and gaps

The triplet gap closes slowly with L_x: from 0.66 to 0.35 at $J_2 = 0$ and from 0.73 to 0.51 at $J_2 = 0.3$ (Fig. 1c). Frustration opens the finite-size gap here. Because the Weyl bound grows as N, the scaled gap $\Delta = g/W$ that sets the filter degree falls roughly as $1/N$. The DMRG energy errors grow from 10^{-10} (4×3, exact at $\chi = 64$) to 2–5×10^{-4} at 4×12. Where ED exists, DMRG matches it to 10^{-11} or better.

3 Trial states: overlap and circuit cost

Compared with the chain, the snake MPS converges much more slowly in χ. The 48-site chain reaches $1 - \gamma \leq 10^{-2}$ at $\chi = 8$; the 4×12 cylinder needs $\chi = 32$ for the same, and $\chi = 64$ leaves 5.5×10^{-4} ($J_2 = 0$). The trial cost grows as $\chi^2 N$, so each doubling of χ costs about $4\times$. That combination (slow convergence, quadratic cost) is what opens room for the filter.

Table 1: Inputs per cylinder. E_0, E_1 from DMRG at $\chi \leq 256$; the errors are the pipeline inputs. $E_{\max}$ is the Weyl upper bound, "time" is the wall-clock DMRG time in hours, and the last column gives $|E^{\mathrm{DMRG}} - E^{\mathrm{ED}}|$ for E_0 and E_1 where ED is available.

system	N	J_2	E_0	E_1	gap	δE_0	δE_1	$E_{\max}$	time [h]	ED check
4×3	12	0	-7.530137	-6.871724	0.6584	1.0×10^{-10}	1.0×10^{-10}	5.0	0.00	2×10^{-14} / 4×10^{-14}
4×4	16	0	-10.264290	-9.688511	0.5758	1.3×10^{-6}	2.2×10^{-6}	7.0	0.12	2×10^{-14} / 9×10^{-12}
4×6	24	0	-15.728450	-15.255513	0.4729	2.3×10^{-5}	3.4×10^{-5}	11.0	1.41	–
4×8	32	0	-21.194186	-20.779712	0.4145	6.3×10^{-5}	1.0×10^{-4}	15.0	3.11	–
4×12	48	0	-32.126439	-31.775997	0.3504	1.9×10^{-4}	3.4×10^{-4}	23.0	6.29	–
4×3	12	0.3	-6.598094	-5.869734	0.7284	1.0×10^{-10}	1.0×10^{-10}	6.2	0.00	2×10^{-14} / 8×10^{-15}
4×4	16	0.3	-8.888556	-8.220986	0.6676	3.4×10^{-6}	2.2×10^{-6}	8.8	0.12	8×10^{-14} / 1×10^{-11}
4×6	24	0.3	-13.465810	-12.874454	0.5914	2.8×10^{-5}	7.8×10^{-5}	14.0	1.46	–
4×8	32	0.3	-18.043497	-17.492736	0.5508	5.9×10^{-5}	2.1×10^{-4}	19.2	2.56	–
4×12	48	0.3	-27.198944	-26.688843	0.5101	1.2×10^{-4}	5.4×10^{-4}	29.6	6.11	–

Table 2: DMRG convergence of E_0: $|E_0(\chi) - E_0(256)|$.

system	J_2	$\chi = 16$	32	64	128	256
4×3	0	2×10^{-2}	7×10^{-4}	8×10^{-15}	1×10^{-14}	ref
4×4	0	5×10^{-2}	4×10^{-3}	1×10^{-4}	1×10^{-6}	ref
4×6	0	9×10^{-2}	1×10^{-2}	5×10^{-4}	2×10^{-5}	ref
4×8	0	1×10^{-1}	2×10^{-2}	1×10^{-3}	6×10^{-5}	ref
4×12	0	2×10^{-1}	4×10^{-2}	2×10^{-3}	2×10^{-4}	ref
4×3	0.3	1×10^{-2}	3×10^{-4}	6×10^{-15}	3×10^{-15}	ref
4×4	0.3	3×10^{-2}	2×10^{-3}	1×10^{-4}	3×10^{-6}	ref
4×6	0.3	5×10^{-2}	6×10^{-3}	5×10^{-4}	3×10^{-5}	ref
4×8	0.3	8×10^{-2}	1×10^{-2}	9×10^{-4}	6×10^{-5}	ref
4×12	0.3	1×10^{-1}	2×10^{-2}	2×10^{-3}	1×10^{-4}	ref

4 Cheapest route to a fixed target

Table 5 compares three routes at $\epsilon = 10^{-2}, 10^{-3}, 10^{-4}$:

- the cheapest MPS that meets $1 - \gamma \leq \epsilon$ on its own;
- the cheapest certified filtered plan (any χ; dyadic, optimized or QETU);
- the cheapest empirical filtered plan (QETU at the degree law, one step per query).

Filtered costs are expected CNOTs per success, including the trial circuit on every attempt.

Certified. The certified filter never beats a compiled MPS that reaches the target, usually missing by one to two orders of magnitude. It is the only route at $\epsilon = 10^{-4}$ on 4×8 and 4×12, where no $\chi \leq 64$ trial suffices. Its excess comes almost entirely from the Trotter bound, which asks for about 16 steps per query where one is enough.

Empirical. With the measured filter cost the comparison depends on where the target falls on the MPS staircase. The empirical filter wins whenever the target lies just beyond what the current χ reaches, so that the MPS would need the next doubling: for example 4×4 at 10^{-2} and 10^{-4}, 4×6 and 4×8 at 10^{-3}, and 4×12 at 10^{-2} ($J_2 = 0$). Otherwise it loses, by up to $6\times$ (e.g. 4×12, $J_2 = 0.3$, 10^{-2}, where $\chi = 32$ already suffices). It is the only route at 10^{-4} on 4×8 and 4×12.

Check of the empirical laws. For 4×3 and 4×4 (and the 16-site chain) the trial state vectors are available. There we ran the actual QETU circuit, one Strang step per query, at the degree the law predicts. In 48 of 48 cases it met the target at exactly that degree (Table 6).

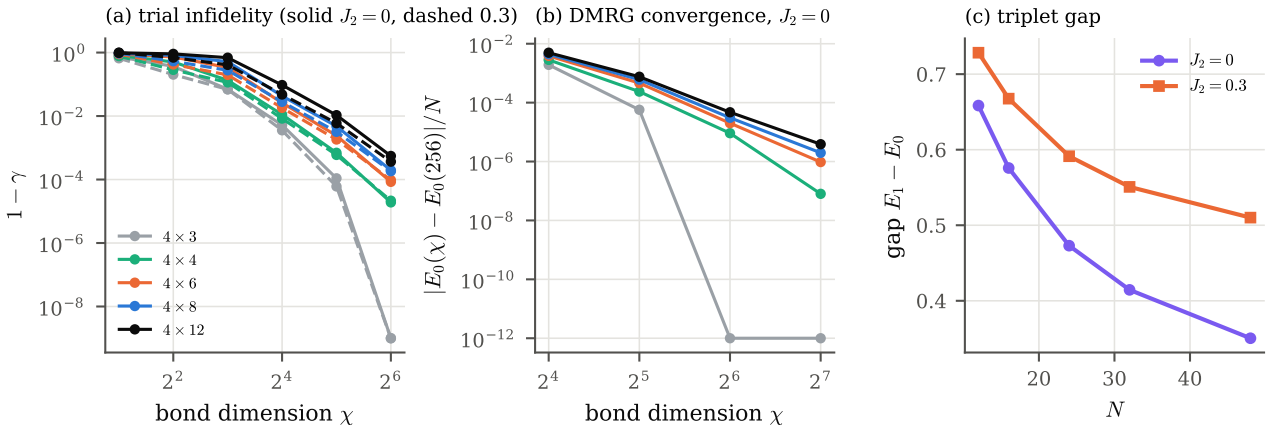


Figure 1: (a) Trial infidelity $1 - \gamma$ versus bond dimension for every cylinder (solid $J_2 = 0$, dashed $J_2 = 0.3$). (b) DMRG energy convergence per site. (c) Triplet gap versus N.

Table 3: Trial infidelity $1 - \gamma$ of the variational MPS. Values at the 10^{-4} level and below for $L_x \geq 8$ are limited by the accuracy of the $\chi = 256$ reference.

system	J_2	$\chi = 2$	4	8	16	32	64
4×3	0	7.3×10^{-1}	3.7×10^{-1}	7.0×10^{-2}	5.2×10^{-3}	1.1×10^{-4}	$< 10^{-9}$
4×4	0	8.1×10^{-1}	5.1×10^{-1}	1.3×10^{-1}	1.1×10^{-2}	7.0×10^{-4}	1.9×10^{-5}
4×6	0	9.0×10^{-1}	7.3×10^{-1}	3.4×10^{-1}	2.6×10^{-2}	2.4×10^{-3}	8.5×10^{-5}
4×8	0	9.5×10^{-1}	8.1×10^{-1}	5.2×10^{-1}	4.3×10^{-2}	4.8×10^{-3}	2.0×10^{-4}
4×12	0	9.8×10^{-1}	9.1×10^{-1}	6.9×10^{-1}	9.5×10^{-2}	1.1×10^{-2}	5.5×10^{-4}
4×3	0.3	6.6×10^{-1}	2.0×10^{-1}	6.9×10^{-2}	3.6×10^{-3}	6.1×10^{-5}	$< 10^{-9}$
4×4	0.3	7.8×10^{-1}	2.9×10^{-1}	1.1×10^{-1}	8.3×10^{-3}	6.0×10^{-4}	2.2×10^{-5}
4×6	0.3	8.9×10^{-1}	4.2×10^{-1}	2.0×10^{-1}	1.8×10^{-2}	1.8×10^{-3}	9.6×10^{-5}
4×8	0.3	9.5×10^{-1}	5.3×10^{-1}	2.8×10^{-1}	2.8×10^{-2}	3.2×10^{-3}	1.8×10^{-4}
4×12	0.3	9.9×10^{-1}	7.0×10^{-1}	4.1×10^{-1}	4.8×10^{-2}	6.0×10^{-3}	3.7×10^{-4}

5 Cost per accuracy and the crossover

For a continuous view we compute the cheapest cost to reach every infidelity between 10^{-1} and 10^{-7} (Fig. 2):

- MPS alone: the cheapest χ with $1 - \gamma \leq \epsilon$, a staircase in ϵ;
- the empirical filter: minimized over the trial χ;
- the certified filter: at the three targets above.

Reading the crossover. On every cylinder the empirical filter wins in the accuracy window just before the MPS needs its next doubling of χ: a short filter on the current χ is cheaper than a 4× larger trial circuit. The saving in that window is a factor of 2–3. Two trends follow from the inputs:

- The window moves to looser accuracy as the cylinder grows (from around 10^{-4} at 4×4 to around 10^{-2} at 4×12). Each χ buys less overlap on a longer cylinder, while the filter degree depends on the overlap only logarithmically.
- Beyond the MPS limit (2×10^{-5} at 4×4, 5×10^{-4} at 4×12) the filter is the only route in our data. A $\chi = 128$ trial would cost about 4× the $\chi = 64$ one, roughly 1.3×10^6 CNOTs at 4×12, about what the empirical filter needs for $1 - F = 10^{-6}$ and more than it needs at any looser target, even if such a trial reached that accuracy.

The chain has no such crossover (see the regime section of the paper): its MPS overlap converges fast enough in χ that the next doubling is always the cheaper option.

Table 4: CNOTs of the MPS preparation circuit (independent of J_2).

system	$\chi = 2$	4	8	16	32	64
4×3	3.3×10^1	2.0×10^2	9.0×10^2	3.6×10^3	1.3×10^4	4.6×10^4
4×4	4.5×10^1	2.8×10^2	1.3×10^3	5.3×10^3	2.1×10^4	7.7×10^4
4×6	6.9×10^1	4.4×10^2	2.1×10^3	8.9×10^3	3.5×10^4	1.4×10^5
4×8	9.3×10^1	6.0×10^2	2.9×10^3	1.2×10^4	5.0×10^4	2.0×10^5
4×12	1.4×10^2	9.2×10^2	4.5×10^3	2.0×10^4	8.0×10^4	3.2×10^5

Table 5: CNOTs (bond dimension in parentheses). M: MPS alone. C: certified filter. E: empirical filter. "–": no $\chi \leq 64$ route.

		$\epsilon = 10^{-2}$			$\epsilon = 10^{-3}$			$\epsilon = 10^{-4}$		
system	J_2	M	C	E	M	C	E	M	C	E
4×3	0	3.6×10^3 (16)	4.9×10^4 (8)	1.2×10^4 (8)	1.3×10^4 (32)	7.4×10^4 (16)	1.2×10^4 (16)	4.6×10^4 (64)	6.5×10^4 (32)	1.5×10^4 (32)
4×4	0	2.1×10^4 (32)	6.3×10^4 (16)	1.1×10^4 (16)	2.1×10^4 (32)	2.7×10^5 (16)	3.0×10^4 (16)	7.7×10^4 (64)	4.0×10^5 (32)	4.2×10^4 (32)
4×6	0	3.5×10^4 (32)	2.6×10^5 (16)	5.2×10^4 (16)	1.4×10^5 (64)	4.3×10^5 (32)	7.6×10^4 (32)	1.4×10^5 (64)	1.6×10^6 (32)	1.2×10^5 (32)
4×8	0	5.0×10^4 (32)	6.9×10^5 (16)	1.3×10^5 (16)	2.0×10^5 (64)	1.2×10^6 (32)	1.7×10^5 (32)	–	1.4×10^6 (64)	2.6×10^5 (32)
4×12	0	3.2×10^5 (64)	7.5×10^5 (32)	1.4×10^5 (32)	3.2×10^5 (64)	3.9×10^6 (32)	5.0×10^5 (32)	–	5.7×10^6 (64)	6.6×10^5 (64)
4×3	0.3	3.6×10^3 (16)	1.2×10^5 (8)	1.9×10^4 (8)	1.3×10^4 (32)	1.3×10^5 (16)	1.6×10^4 (16)	1.3×10^4 (32)	4.5×10^5 (16)	2.8×10^4 (16)
4×4	0.3	5.3×10^3 (16)	4.8×10^5 (8)	4.9×10^4 (8)	2.1×10^4 (32)	5.6×10^5 (16)	4.3×10^4 (16)	7.7×10^4 (64)	9.1×10^5 (32)	5.3×10^4 (32)
4×6	0.3	3.5×10^4 (32)	5.0×10^5 (16)	6.2×10^4 (16)	1.4×10^5 (64)	8.2×10^5 (32)	8.8×10^4 (32)	1.4×10^5 (64)	3.4×10^6 (32)	1.6×10^5 (32)
4×8	0.3	5.0×10^4 (32)	1.2×10^6 (16)	1.6×10^5 (16)	2.0×10^5 (64)	2.0×10^6 (32)	2.0×10^5 (32)	–	2.7×10^6 (64)	3.0×10^5 (64)
4×12	0.3	8.0×10^4 (32)	3.3×10^6 (16)	4.9×10^5 (16)	3.2×10^5 (64)	5.9×10^6 (32)	5.6×10^5 (32)	–	8.5×10^6 (64)	7.1×10^5 (64)

6 Caveats

- The empirical filter cost rests on two laws measured for $N \leq 16$ (chain and the two small cylinders). They held in every case we could simulate, but beyond $N = 16$ they are extrapolated, and they are not a certificate. Only the certified curve carries a guarantee, and it does not cross the MPS.
- Overlaps at the 10^{-4} level and below for $L_x \geq 8$, and energy errors of 2–5 $\times 10^{-4}$ at 4×12, are limited by the $\chi = 256$ DMRG reference. The MPS staircase is therefore uncertain at its tight end on the larger cylinders.
- ED confirms the triplet as the first excitation only for 4×3 and 4×4. For larger cylinders this is an assumption.
- Costs are CNOT counts with all-to-all connectivity. Trial circuits use a generic decomposition bound, not compiled circuits. Repeat-until-success is assumed throughout.
- Only width-4 cylinders were studied. Wider cylinders, where the snake MPS degrades faster, are the natural next test of whether the crossover window widens.

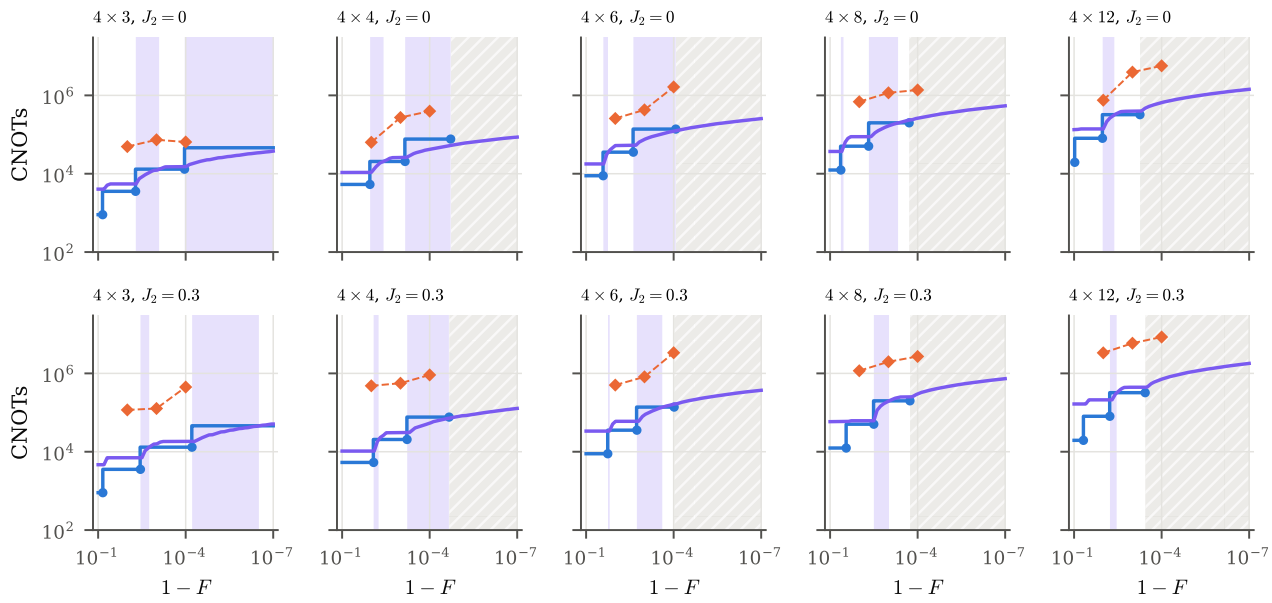


Figure 2: Cheapest CNOTs to reach infidelity $1 - F$ (accuracy increases to the right). Blue: MPS alone (dots are the $\chi = 2, \ldots, 64$ trials). Purple: MPS + filter, empirical. Orange diamonds: MPS + filter, certified. Purple shading: the empirical filter is cheaper. Grey hatching: no MPS with $\chi \leq 64$ reaches that accuracy.

Table 6: State-vector simulation of the empirical filter on cylinders: predicted degree, degree that reached ϵ with one Strang step per query, measured infidelity and success probability.

system	J_2	χ	γ	ϵ	d (law)	d (sim)	$1-F$	p
4×3	0	2	0.2697	1×10^{-2}	44	44	4.5×10^{-3}	0.271
4×3	0.3	2	0.3391	1×10^{-2}	38	38	5.0×10^{-3}	0.340
4×3	0	2	0.2697	1×10^{-3}	58	58	3.9×10^{-4}	0.270
4×3	0.3	2	0.3391	1×10^{-3}	52	52	3.6×10^{-4}	0.338
4×3	0	2	0.2697	1×10^{-4}	72	72	4.9×10^{-5}	0.269
4×3	0.3	2	0.3391	1×10^{-4}	64	64	3.4×10^{-5}	0.338
4×3	0	4	0.6333	1×10^{-2}	34	34	5.0×10^{-3}	0.638
4×3	0.3	4	0.7978	1×10^{-2}	26	26	2.6×10^{-3}	0.800
4×3	0	4	0.6333	1×10^{-3}	48	48	4.5×10^{-4}	0.634
4×3	0.3	4	0.7978	1×10^{-3}	40	40	5.7×10^{-4}	0.798
4×3	0	4	0.6333	1×10^{-4}	62	62	9.0×10^{-5}	0.634
4×3	0.3	4	0.7978	1×10^{-4}	52	52	3.1×10^{-5}	0.797
4×3	0	8	0.9300	1×10^{-2}	22	22	5.7×10^{-3}	0.936
4×3	0.3	8	0.9309	1×10^{-2}	20	20	6.0×10^{-3}	0.936
4×3	0	8	0.9300	1×10^{-3}	36	36	3.2×10^{-4}	0.931
4×3	0.3	8	0.9309	1×10^{-3}	32	32	5.2×10^{-4}	0.931
4×3	0	8	0.9300	1×10^{-4}	50	50	3.2×10^{-5}	0.930
4×3	0.3	8	0.9309	1×10^{-4}	46	46	3.9×10^{-5}	0.930
4×3	0	16	0.9948	1×10^{-3}	18	18	5.2×10^{-4}	0.995
4×3	0.3	16	0.9964	1×10^{-3}	14	14	3.0×10^{-4}	0.996
4×3	0	16	0.9948	1×10^{-4}	34	34	4.1×10^{-5}	0.994
4×3	0.3	16	0.9964	1×10^{-4}	28	28	5.5×10^{-5}	0.995
4×3	0	32	0.9999	1×10^{-4}	4	4	9.2×10^{-5}	1.000
4×4	0	2	0.1891	1×10^{-2}	72	72	4.9×10^{-3}	0.190
4×4	0.3	2	0.2249	1×10^{-2}	62	62	4.5×10^{-3}	0.225
4×4	0	2	0.1891	1×10^{-3}	94	94	4.6×10^{-4}	0.189
4×4	0.3	2	0.2249	1×10^{-3}	82	82	3.9×10^{-4}	0.224
4×4	0	2	0.1891	1×10^{-4}	116	116	5.0×10^{-5}	0.189
4×4	0.3	2	0.2249	1×10^{-4}	100	100	4.1×10^{-5}	0.224
4×4	0	4	0.4946	1×10^{-2}	58	58	5.4×10^{-3}	0.498
4×4	0.3	4	0.7148	1×10^{-2}	44	44	2.7×10^{-3}	0.716
4×4	0	4	0.4946	1×10^{-3}	80	80	4.2×10^{-4}	0.495
4×4	0.3	4	0.7148	1×10^{-3}	64	64	3.0×10^{-4}	0.715
4×4	0	4	0.4946	1×10^{-4}	102	102	4.2×10^{-5}	0.495
4×4	0.3	4	0.7148	1×10^{-4}	82	82	6.4×10^{-5}	0.714
4×4	0	8	0.8660	1×10^{-2}	40	40	5.5×10^{-3}	0.871
4×4	0.3	8	0.8852	1×10^{-2}	34	34	5.6×10^{-3}	0.890
4×4	0	8	0.8660	1×10^{-3}	62	62	6.2×10^{-4}	0.867
4×4	0.3	8	0.8852	1×10^{-3}	54	54	5.6×10^{-4}	0.885
4×4	0	8	0.8660	1×10^{-4}	84	84	7.0×10^{-5}	0.866
4×4	0.3	8	0.8852	1×10^{-4}	74	74	4.0×10^{-5}	0.884
4×4	0	16	0.9889	1×10^{-2}	8	8	3.9×10^{-3}	0.993
4×4	0	16	0.9889	1×10^{-3}	36	36	5.6×10^{-4}	0.989
4×4	0.3	16	0.9917	1×10^{-3}	30	30	5.7×10^{-4}	0.992
4×4	0	16	0.9889	1×10^{-4}	60	60	4.3×10^{-5}	0.988
4×4	0.3	16	0.9917	1×10^{-4}	50	50	5.3×10^{-5}	0.991
4×4	0	32	0.9993	1×10^{-4}	32	32	7.0×10^{-5}	0.999
4×4	0.3	32	0.9994	1×10^{-4}	26	26	5.1×10^{-5}	0.999

Table 7: Accuracy windows in which the empirical filter is cheaper than any MPS that meets the target alone (largest saving in parentheses; windows narrower than a factor 1.5 in $1 - F$ omitted), and the accuracy beyond which no $\chi \leq 64$ MPS suffices.

system	J_2	filter cheaper for $1 - F$ in	MPS ($\chi \leq 64$) limit
4×3	0	5×10^{-3}–8×10^{-4} (2.4×); 1×10^{-4}–1×10^{-7} (3.1×)	–
4×4	0	1×10^{-2}–4×10^{-3} (1.9×); 7×10^{-4}–2×10^{-5} (3.0×)	2×10^{-5}
4×6	0	2×10^{-3}–9×10^{-5} (2.6×)	8×10^{-5}
4×8	0	5×10^{-3}–5×10^{-4} (2.3×)	2×10^{-4}
4×12	0	1×10^{-2}–4×10^{-3} (2.3×)	5×10^{-4}
4×3	0.3	4×10^{-3}–2×10^{-3} (1.9×); 6×10^{-5}–3×10^{-7} (2.5×)	–
4×4	0.3	6×10^{-4}–2×10^{-5} (2.5×)	2×10^{-5}
4×6	0.3	2×10^{-3}–2×10^{-4} (2.3×)	1×10^{-4}
4×8	0.3	3×10^{-3}–1×10^{-3} (3.2×)	2×10^{-4}
4×12	0.3	6×10^{-3}–3×10^{-3} (1.5×)	4×10^{-4}